

Artificial Intelligence Concepts -- Guided Notes ANSWER KEY

Unit tp-4.6 | CompTIA Network+ N10-009 Obj. FC0-U71 4.6 | N10-008 FC0-U71 4.6

For instructor use / distribute after students complete the notes.

Question 1

AI (Artificial Intelligence) refers to computer systems designed to perform tasks that normally require human _____, such as recognizing speech, identifying images, and making decisions. _____

Learning is a subset of AI where systems learn from _____ without being explicitly programmed for each task.

Answer: Intelligence; machine; data.

Question 2

In _____ learning, the training data includes labeled examples -- the model learns from input-output pairs. In _____ learning, the model finds patterns in unlabeled data with no predefined answers.

_____ learning trains an agent to take actions in an environment to maximize a reward signal.

Answer: Supervised; unsupervised; reinforcement.

Question 3

A _____ neural network is designed to recognize patterns in image data by scanning the image in overlapping regions. It is used in facial recognition, medical imaging, and self-driving cars. _____

neural networks (RNNs) are designed for _____ data such as text and audio because they retain context from previous inputs.

Answer: Convolutional (CNN); recurrent; sequential (time-series).

Question 4

A _____ model (like GPT) predicts the next word in a sequence based on billions of parameters trained on large text datasets. These are also called _____ Models (LLMs). They can generate human-like text, translate languages, write code, and answer questions. A key limitation is that they occasionally produce confident but _____ outputs, called _____.

Answer: Language; Large Language; incorrect (false); hallucinations.

Question 5

_____ AI refers to systems that excel at one specific narrow task (playing chess, recognizing faces).

_____ AI (AGI) would perform any intellectual task a human can -- it does not yet exist. AI _____ refers to unintended harmful outcomes when AI systems reflect biases present in _____ data.

Answer: Narrow; artificial general; bias; training.

Question 6

AI _____ is the ability to explain why an AI model made a specific decision. This is important in high-stakes domains like _____, _____, and _____ where decisions must be justifiable to

affected individuals. _____ networks are often called black boxes because their decision processes are hard to interpret.

Answer: Explainability (interpretability); healthcare, lending/credit, law enforcement (any two); neural.

Question 7

Generative AI creates new _____ -- text, images, audio, video, or code -- based on patterns learned from training data. Tools like _____ generate images from text prompts, while _____ generates text. These tools raise ethical concerns about _____ (generating fake but realistic media) and _____.

Answer: Content; DALL-E / Midjourney; ChatGPT (examples); deepfakes; copyright.

Question 8 -- Real World Application

REAL WORLD: A school uses an AI tool to screen student applications and rank them. A student advocacy group finds that the model rejects more applications from students at lower-income zip codes, even when their grades are equal to students from higher-income zip codes. Explain what AI problem is occurring and recommend two steps the school should take.

Answer: The problem is AI bias -- specifically training data bias. If the model was trained on historical admissions data that reflected socioeconomic inequality (e.g., the school historically admitted more students from wealthy zip codes), the model has learned to replicate that discrimination. It has encoded historical bias as a pattern to follow, not a mistake to correct. Step 1 -- Audit the model for disparate impact: analyze the model decisions disaggregated by demographic groups (income, zip code, race). If acceptance rates differ significantly for equally qualified students from different groups, the model is discriminating. Use statistical tests to quantify the disparity. Step 2 -- Mitigate bias and add human oversight: retrain the model with bias-mitigating techniques (resampling, re-weighting underrepresented groups) and exclude or carefully handle proxies for protected characteristics like zip code. Most importantly, AI should not be the final decision-maker for consequential decisions about individuals -- add mandatory human review before any rejection.